Tharwat Farag Badreldin

DATA ENGINEER | SQL & DATABASE ENGINEERING | RETAIL DATA PLATFORMS

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LinkedIn | GitHub

Professional Summary

Data Engineer with professional experience in SQL, database administration, technical troubleshooting, and client-facing platform support, complemented by hands-on Data Engineering projects. Completed the Digilians Data Engineering Program at the Egyptian Military Academy in collaboration with MCIT, applying Python, ETL/ELT, data-quality controls, Microsoft Fabric, and dimensional modeling in practical projects. Previously led technical delivery and post-launch support for 100+ LMS/EdTech platforms serving 130,000+ active users, working with clients and technical teams to resolve issues and maintain reliable systems. Combines database expertise, customer-focused problem solving, and practical pipeline development to help organizations build dependable, analytics-ready data.

Technical Skills

Programming & Querying: Python, SQL, Pandas, NumPy

Data Engineering: ETL/ELT Pipelines, Data Ingestion, Data Transformation, Data Quality Validation, Deduplication, Quarantine Handling, Data Reconciliation

Data Warehousing & Modeling: Star Schema, Fact and Dimension Tables, Relational Modeling, Data Warehouse Design, Analytical SQL, SCD Type 1/2

Databases: PostgreSQL, SQL Server, MySQL, SQLite; Database Administration, Query Optimization, Indexing, Data Integrity, Production Troubleshooting

Platforms & Orchestration: Microsoft Fabric (Data Factory, Lakehouse, OneLake, Warehouse, Notebooks), SSIS, Apache Airflow, Docker

Big Data & Cloud: PySpark / Apache Spark, Hadoop, Hive, Microsoft Azure

Development Tools: Git, GitHub, pytest, Jupyter Notebook, Visual Studio Code

Professional Skills: Technical Leadership, Client Follow-up, Post-delivery Support, Requirements Gathering, Stakeholder Communication, Cross-functional Collaboration

Professional Experience & Data Engineering Training

Data Engineering Specialist Trainee
Jan 2026 – Jun 2026

Digilians – Egyptian Military Academy, in collaboration with MCIT — *Cairo, Egypt*

- Completed hands-on training exercises in Python and SQL covering data extraction, transformation, validation, and loading for analytical use cases.
- Practiced SQL query optimization, indexing concepts, and relational data modeling through technical exercises and data engineering projects.
- Gained practical exposure to Microsoft Fabric, Azure data services, ETL/ELT concepts, data warehousing, and pipeline design.

Technical Lead – LMS & EdTech Platforms
Jul 2020 – Dec 2025

Camp Coding — *Tanta, Egypt*

- Led technical delivery and ongoing client support for 100+ LMS/EdTech platforms serving 130,000+ active users, supporting customers beyond launch and handover.
- Followed up directly with clients on post-delivery issues, clarified their operational needs, coordinated troubleshooting, and stayed engaged until reported problems were addressed.
- Investigated production incidents across SQL databases, application behavior, configuration, and backend-related issues; used database administration, data validation, and record-consistency checks to identify root causes.
- Worked between clients and internal technical teams to turn real-world user problems into clear technical actions, prioritize fixes, and help keep business-critical platforms reliable.

IT Manager
Nov 2023 – Jul 2025

Baseeta — *Part-time*

- Oversaw IT operations and website performance initiatives, coordinating technical priorities with business stakeholders.

Software Sales Specialist
Dec 2024 – Jun 2025

Alnaierh — *Riyadh, Saudi Arabia – Remote*

- Coordinated communication between business stakeholders and technical teams to support software operations and website lifecycle activities.

Selected Data Engineering Projects

RetailFlow Supply Chain & Inventory Data Engineering Pipeline GitHub

- Built an end-to-end supply chain data pipeline using Python, PostgreSQL, and Docker, processing raw data through Bronze and Silver layers into a dimensional analytics warehouse.
- Implemented schema contracts, data-quality validation, deterministic deduplication, quarantine handling, and SHA-256 ingestion manifests to support source traceability.
- Designed a Star Schema for inventory snapshots and purchase-order fulfillment, with transactional, idempotent loading and a documented 55-test automated suite.

B.Laban Retail Customer & Sales Analytics Platform GitHub

- Built a retail analytics pipeline using PySpark, Microsoft Fabric, Delta Lake, and SQL with a Bronze → Silver → Gold medallion architecture.
- Implemented data-quality checks, standardization, anomaly flagging, and automated tests across 7K transactional records, identifying 48 invalid transactions while preserving source traceability.
- Prepared Gold datasets for branch, product, channel, customer, and daily-sales analysis to support downstream reporting.

SSIS Data Warehouse ETL Pipeline GitHub

- Built a SQL Server data warehouse using SSIS and Star Schema modeling, implementing ETL workflows with SCD Type 1/2 handling and SSAS/SSRS reporting.

Apache Airflow ETL Pipeline GitHub

- Developed automated ETL workflows using Apache Airflow and Astronomer in a Dockerized environment.

Real Estate Data Platform GitHub

- Developed a Python and Pandas pipeline using web-scraped property listings, SQL Server, data cleaning, quality checks, and analytical SQL views for sale and rental KPIs.

Education

B.Sc. in Computer Science **Sep 2019 – Jun 2023**

Faculty of Computers and Information, Tanta University *Tanta, Egypt*

- Very Good Grade

Certifications

Microsoft Certified: Fabric Data Engineer Associate (DP-700)

Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)

Microsoft Certified: SQL AI Developer Associate (DP-800)

Microsoft Certified: Azure Data Fundamentals (DP-900)

Languages

Arabic: Native | **English:** Professional Working Proficiency